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MilaJovanoska

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Profile

MSc student in **Robotics at the University of Manchester**, holding a **BSc in Software Engineering and Information Systems (240 ECTS)** from the Faculty of Computer Science and Engineering (FINKI), with a **final GPA of 9.82/10.00**. My academic interests focus on **Robotics, Artificial Intelligence, Machine Learning, and Software Engineering**, supported by a strong foundation in mathematics, statistics, and programming. I have experience in **machine learning research**, including a conference paper on tuberculosis incidence prediction, alongside developing **full-stack applications and CI/CD workflows**. I have also **led multiple team projects** and gained valuable **teaching and mentoring experience** through undergraduate laboratory instruction.

Education

2026–Present **MSc in Robotics**, *The University of Manchester*, Manchester, United Kingdom
Expected graduation: **2027**.
Relevant courses: Robotic Systems; Software for Robotics; Robotic Systems Design Project; Cognitive Robotics and Computer Vision.

2022–2026 **BSc in Software Engineering and Information Systems (240 ECTS)**, *Faculty of Computer Science and Engineering (FINKI)*, Ss. Cyril and Methodius University, Skopje, North Macedonia
Final GPA: 9.82/10.00.
Relevant courses: Machine Learning, Natural Language Processing, Linear Algebra, Probability and Statistics, Statistical Modelling, Algorithms and Data Structures, Databases, Software Engineering.

2018–2022 **High School Diploma – Natural Sciences and Mathematics (PMA)**, *Orce Nikolov High School*, Skopje, North Macedonia
GPA: 5.0/5.0.

Undergraduate Dissertation

Jan 2026 – Jul 2026 **Statistically Supported and Interpretable Classification of Countries Based on Tuberculosis Burden**, *BSc Dissertation, Software Engineering and Information Systems (FINKI)*, Individual Research Project

- Developed a binary classification framework to identify countries with analytically defined higher tuberculosis burden, combining nutrition, socioeconomic, human development and vaccination indicators across **183 countries**.
- Conducted statistically supported feature selection using **Welch's t-test, Mann–Whitney U tests, Benjamini–Hochberg false discovery rate correction and effect sizes**; assessed dependence and multicollinearity using Spearman correlation and variance inflation factors.
- Evaluated **Logistic Regression** models using full and selected feature sets and examined feature contributions with **permutation importance**, distinguishing statistical associations from causal conclusions.

Publications

Apr 2026 **Predicting Tuberculosis Incidence in 2022: A Ridge Regression and SHAP Analysis**, *Mila Jovanoska, Andrej Ristovski, Anika Ristevska, Slobodan Kalajdziski, Proceedings of the 23rd International Conference on Informatics and Information Technologies (CIIT 2026)*
Struga, North Macedonia, 24–26 April 2026.
Publication: CIIT 2026 Proceedings (paper starts on p. 181).

Projects

May 2026 – Jun 2026 **Recipe Manager – Full-Stack Application and CI/CD Workflow**, *Course Project (Continuous Integration and Delivery, FINKI) | Individual*

- Developed a full-stack recipe management application using **React, TypeScript, Spring Boot and PostgreSQL**, with database migrations managed through **Flyway**.
- Containerised application services using **Docker and Docker Compose**, including multi-stage builds to support consistent local deployment.
- Configured **GitHub Actions** for backend tests, frontend checks, and Docker image builds and publication to Docker Hub; prepared **Kubernetes manifests** for deployment configuration.
- **Code:** github.com/MilaJovanoska/recipe-manager-ci-cd

May 2025 – Jul 2025 **Predicting Alzheimer's Disease Using Machine Learning: A Classification Approach**, *Course Project (Machine Learning, FINKI) | Individual*

- Built a binary classification pipeline to predict Alzheimer's diagnosis from clinical, demographic and lifestyle variables in a dataset of **2,149 patient records**.
- Applied preprocessing, scaling and class-imbalance handling with **SMOTE**; explored feature selection using Mutual Information, recursive feature elimination and Random Forest feature importance.
- Compared **Logistic Regression, SVM, kNN, LDA/QDA, Random Forest and MLP** models. Exploratory results reached **95% accuracy** and **ROC–AUC of 0.942**; these results require cautious interpretation because of potential data leakage.
- **Code:** github.com/MilaJovanoska/alzheimers-disease-prediction-ml

May 2025 – Sep 2026 **Modelling Extreme Tsunami Economic Damages with the Weibull Distribution (EM-DAT)**, *Course Project (Statistical Modelling, FINKI) | Individual*

- Cleaned and validated **31 country-level tsunami damage records** from EM-DAT; investigated heavy-tailed behaviour using descriptive and tail-focused statistics.
- Examined temporal and geographical patterns through trend analysis and geospatial visualisations.
- Fitted the **Weibull distribution by maximum likelihood estimation**, constructed bootstrap confidence intervals, and assessed fit using P–P/Q–Q plots, Kolmogorov–Smirnov and Cramér–von Mises tests; compared Weibull, lognormal and gamma models using AIC/BIC.
- **Code:** github.com/MilaJovanoska/weibull-extreme-tsunami-damages

Jun 2025 – Aug 2026 **Radio Signal Strength Prediction System (Digital TV and FM)**, *Course Project (Integrated Systems, FINKI) | Team of 3 (**Team Lead**)*

- Developed supervised regression models for **Digital TV and FM electric field strength** prediction using geospatial and contextual features; integrated Python models into a **C#/.NET** system.
- Designed preprocessing workflows with DMS-to-decimal coordinate conversion, feature engineering and `ColumnTransformer` pipelines.
- Tuned Decision Tree and Random Forest regressors using **GroupKFold** and **RandomizedSearchCV**; saved trained models with `joblib` and implemented model inference through **FastAPI**.
- **Code:** github.com/ManuelTrajcev/RadioSignalsAnalysis

Nov 2025 – Feb 2026 **Food Delivery App (Wolt/Korpa-inspired) – Full-Stack Web Application**, *Course Project (Design of Human–Computer Interaction, FINKI) | Team of 2 (**Team Lead**)*

- Developed a **Django** food-delivery application with modular architecture and role-based access control for customers, restaurant owners, couriers and administrators.
- Implemented restaurant search, shopping basket, checkout, order-status tracking and role-specific dashboards.
- Integrated **Stripe payments in test mode** and simulated courier tracking with **Leaflet/OpenStreetMap and AJAX**; added reproducible sample-data seeding.
- **Code:** github.com/andovskaana/Food-Delivery-App-DNICK

Experience

Mar 2025 – Jul 2025 **Database Engineering Intern**, *Inteligenta.io*, Skopje (Hybrid/On-site)

- **Designed and implemented** a database-level multilingual strategy for a primary-school library management system, avoiding separate columns for each language.
- **Refactored database schemas and queries** using translation tables and keyed localisation entities while preserving referential integrity.
- Supported **requirements analysis, migration planning and validation** using representative library data and usage scenarios.

2024–2025 **Laboratory Teaching Assistant (Lab Instructor)**, *Faculty of Computer Science and Engineering (FINKI)*, Skopje, North Macedonia

- Delivered and supported undergraduate laboratory sessions in **Algorithms and Data Structures** and **Probability and Statistics**.
- Designed weekly exercises and problem sets aligned with course content and learning objectives.
- Provided **student mentoring**, concept explanations, debugging guidance and assistance with teaching materials.

Skills

Programming C, C++, Java, Python, SQL, JavaScript, TypeScript

ML/Data pandas, NumPy, SciPy, scikit-learn, imbalanced-learn (SMOTE), XGBoost, TensorFlow/Keras; feature selection, cross-validation, model evaluation

Web HTML, CSS, Bootstrap, Django, Spring Boot, React, .NET, FastAPI, Thymeleaf

Data/ Visualisation Matplotlib, seaborn, Plotly, GeoPandas

Tools Git, GitHub, Google Colab, Jupyter Notebook, VS Code, PyCharm, IntelliJ IDEA, WebStorm

Robotics ROS 2 (Jazzy), rclpy, RViz2, Ubuntu Linux

DevOps GitHub Actions, Docker, Docker Compose, Kubernetes

Languages

Macedonian Native

English C1 (CEFR); IELTS Academic overall band score: 7.0

French B1 (CEFR)

Awards & Scholarships

2026/2027 Postgraduate Scholarship, Ministry of Education and Science of North Macedonia, supporting MSc Robotics studies at the University of Manchester.

2025/2026 Faculty Excellence Award (FINKI) for students with GPA > 9.5.

2024/2025 Faculty Excellence Award (FINKI) for students with GPA > 9.5.

2023/2024 Faculty Excellence Award (FINKI) for students with GPA > 9.5.

2025/2026 Merit Scholarship (Ministry of Education and Science) for talented students (GPA > 8.5).

2024/2025 Merit Scholarship (Ministry of Education and Science) for talented students (GPA > 8.5).

2023/2024 Merit Scholarship (Ministry of Education and Science) for talented students (GPA > 8.5).